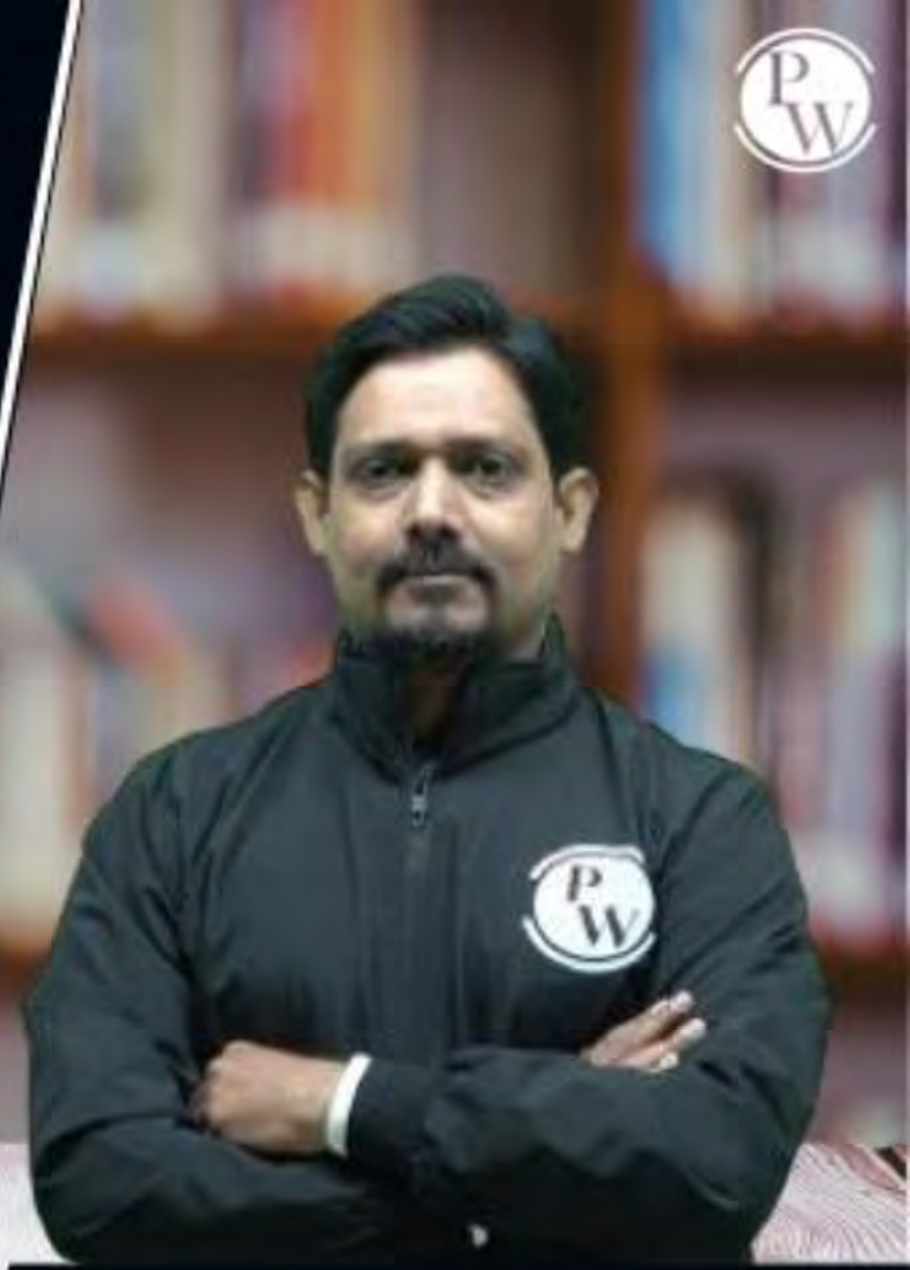
Computer Science & IT



Data Structure & Programming

Stack

Lecture No. 01



By- Abhishek Sir

Recap of Previous Lecture



- Topic: 3-D matrix
- Topic: Tridiagonal Matrix
- Topic: Practice Problem
- Topic
- Topic

Slide

Topics to be Covered



- Topic — Stack as ADT
- Topic — Structure in C Language
- Topic — Practice Problem
- Topic
- Topic

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Topic : Question

9 9

#Q Consider the LTM a[-4..4][0...8] Non zero elements are stored in row major order. What is the address of A[0][3]. Base address is 1000 and size of each element is 2B.

55 56

$BA + \left(\frac{i(i-1)}{2} + j - 1\right) \times s$

attempt

A[1..n][1....n]

A[-4,4][0..8] → 9×9

+5 , +1

A[1..9][1..9]

A[0][3] ← A[5][4]

-4 0
-3 0 0
-2 0 0 0
-1 0 0 0 0
0 1 2 } 3

$1000 + \left[\left(\frac{5 \times 4}{2}\right) + 3\right] \times 2 = 1000 + 13 \times 2 = 1026$ ✓

Slide



Topic : Question

#Q Assume that Tridiagonal Matrix of order $(n \times n)$is mopped into a one dimensional array using column major order, then calculate the location of an element a(i,j)(L0 is base address and size is 1 Byte)?

(A) $L_0+(2j+i-3)$

(B) $L_0+(2i+j-3)$

(C) $L_0+(2i-j-3)$

(D) $L_0+(2j+i+3)$

$i-j+1$

0

$3-2+1=2$

a_{11} a_{12}

a_{21} a_{22}

a_{32}

$3\times(j-1)-1+i-j+1$

$3j-3+i-j = (2j+i-3)\times Size$

Slide



Topic : Stack as LIFO

Linear Data structure

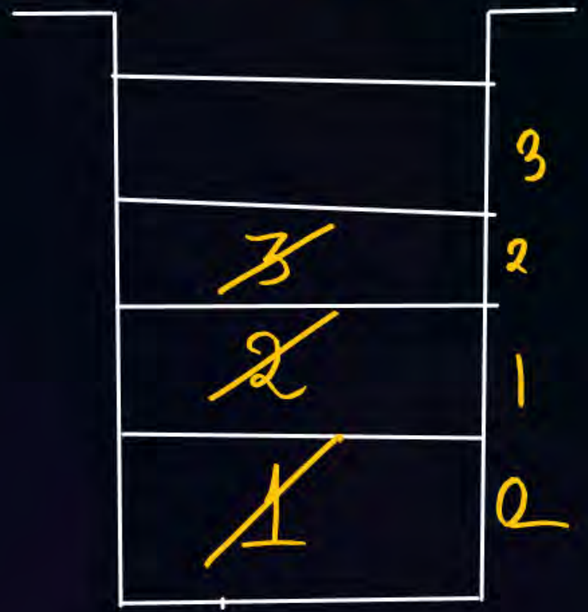
* Stack is Linear Data structure
* Stack works on principle of Last in first out.
* Stack operation controlled an Index (pointer)
  top (stack pointer). points to top of element
* Insertion in stack is called push operation (Index pointed by top)

Slide



Topic : Stack as LIFO

* Deletion from stack is called pop (Deletes element at top of stack)
* Stack is one ended data structure.



push(1) , push(2) , push(3)

pop() - return 3
pop() - return 2
pop() - return 1

Slide



Topic : Stack as LIFO

* Suppose stack size is 5



Stack :

Situation stack empty

IsEmpty() returns True

if stack is empty ✓

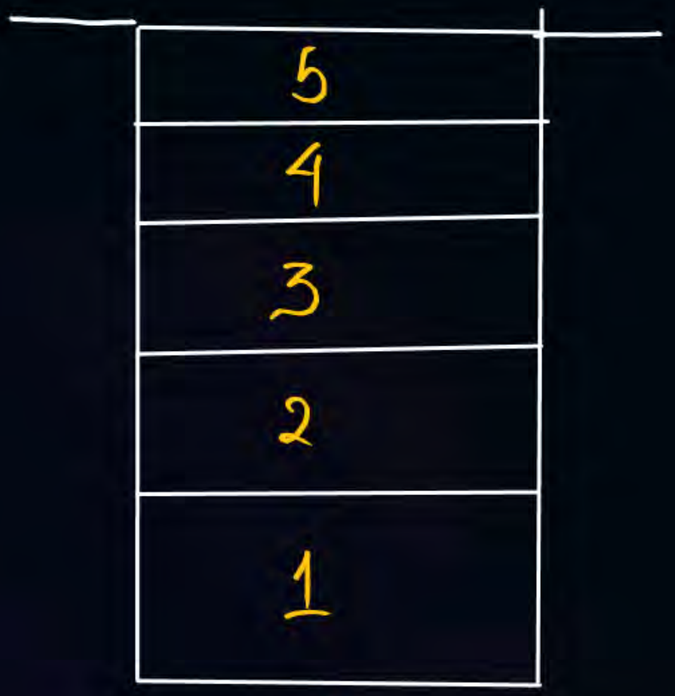
else return false

Slide



Topic : Stack as LIFO

* Suppose stack size is 5



Stack :

push(1), push(2) push(3) push(4)
push(5)

push(6) — what will happen?

Stack full
push(6) → Stack overflow ✓

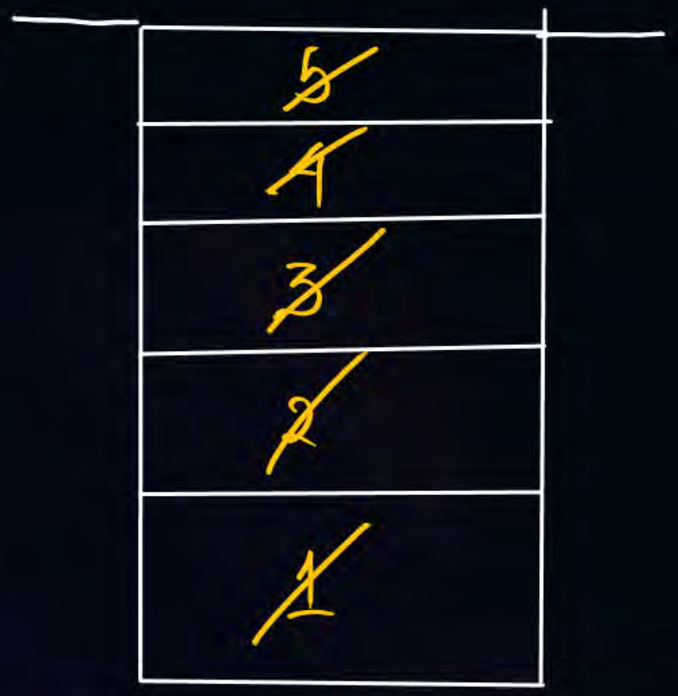
Recursion : Stack overflow

Slide



Topic : Stack as LIFO

* Suppose stack size is 5



Stack :

pop() – return 5

pop() – return 4

pop() – return 3

pop() – return 2

pop() – return 1

pop()

Stack becomes empty

Stack underflow

Slide



Topic : Stack as LIFO

* element Inserted in full stack leads stack overflow
* Element Deleted from empty stack leads to stack underflow.

Application : 1. Implementation of function call.

2. Recursion : Recursion Implemented using Run time stack
3. Divide & Conquer : Recursive
4. Graph traversal. Depth first Search

Slide



Topic : Stack as LIFO

* Expression evaluation - Infix, prefix postfix
* stack as permutation generator.
* Push down Automata (FSM + stack)
* Balanced parethesization (stack)

(((( ))))

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20, 20, 10, 10, 20



20, 20, 10, 10, 20

Q 1991

The following sequence of operation performed on stack.

push(10) push(20). pop(), push(10), push(20)

pop(), pop(), pop() push(20) pop() the Sequence of value popped out

(A) 20 10 20, 10, 20 (B) 20, 20, 10, 10, 20

(C) 10, 20, 20, 10, 20 (D) 20, 20, 10, 20, 10



Topic : Stack

C index starts from 0

```
int a[100];
int top;
```
} global variable

```
void Init () {
    top = -1;
}
```

No value return

```
int IsEmpty () {
    if (top == -1)
        return 1;
    else
        return 0;
}
```

Integer

compare equal to -1

true

false

Slide



Topic : Stack Application

if top equal to -1 then stack is empty

C Index start from '0'

value will filled from 0

0 — 99

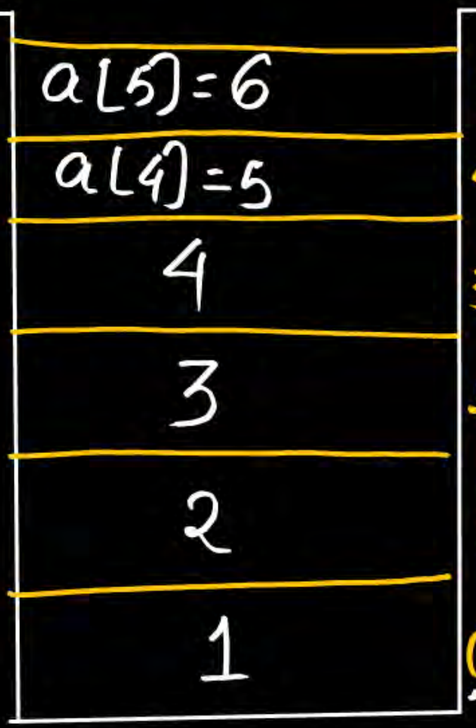
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```
void push (int x) {
    if (top == 99) {
        printf ("stack overflow");
        return;
    }
    top = top + 1;
    a[top] = x;
}
```

Space is available

Increment top

put the element



a[ ]

top = -1

top = ∅ 1 2 3 4 5

push(1)

push(2)

push(3)

push(4) , push(5), push(6)

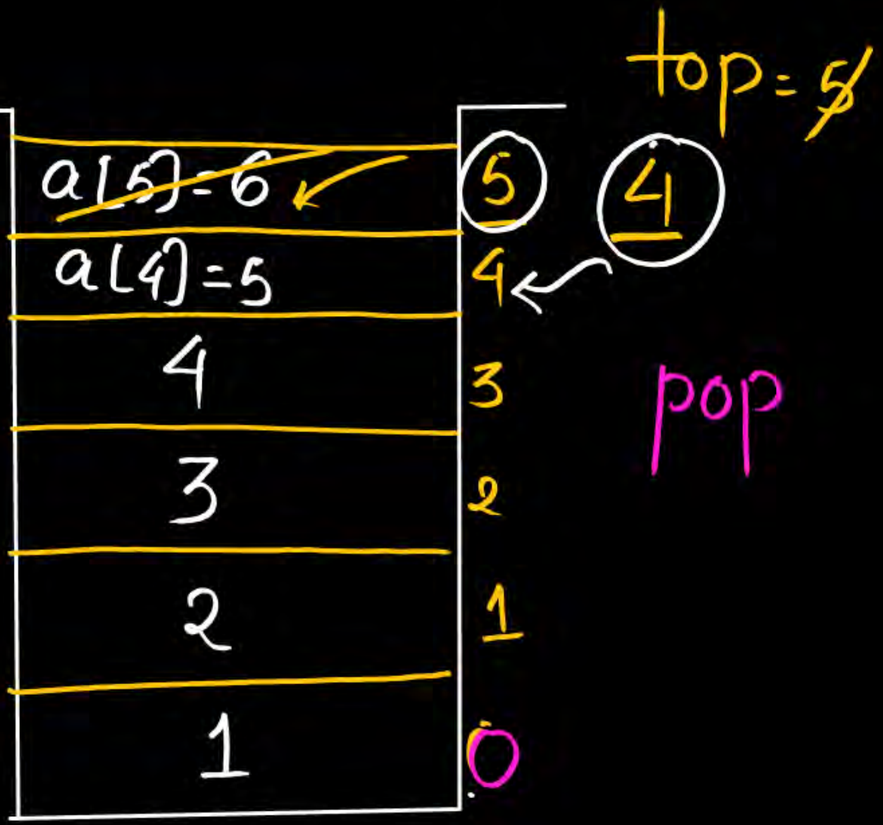
push(7)

return an integer

```
int pop(){
    int data;
    if(top==-1){
        printf("Stack underflow");
        return -1;
    }
    data = a[top];
    top = top - 1;
    return data;
}
```

first take the value

decrement the top





Topic : Stack Application

* if top equal to -1 then stack is empty
* C Index start & from '0'
* push : Increment the top & put value at a[top]
* pop : take element from a[top] and then decrement the top.

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Topic : Stack Application

H.W

(i) The value of Fibonacci (7) is _____

(ii) The total number of calls in Fibonacci(8) is _____

(iii) The total no. of '+' in Fibonacci (9) is_____

```
int fib(int n){
  if(n==0||n==1)
    return n;
  else
    return
      fib(n-1)+fib(n-2);
}
```



2 mins Summary

Topic — Stack (LIFO)

Topic — Init, IsEmpty

Topic — push()

Topic — pop()

Topic

Slide

THANK - YOU

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